STATISTICS · UNIT 1: EXPLORING DATA

Lunch Line Data Detectives

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students act as data analysts for their own cafeteria. They compare two serving lines with displays and summary statistics, check for outliers, use a normal model for daily averages, study lunch choice by grade in a two-way table, and model wait time with least-squares regression.

The project ends with a short data report that recommends one change to the principal.

Standards Addressed

- S-ID.A.1–3 Represent and compare distributions (shape, center, spread, outliers)
- S-ID.A.4 Normal distributions, z-scores, and percents
- S-ID.B.5 Two-way tables and conditional relative frequencies
- S-ID.B.6, S-ID.C.7–9 Scatterplots, least-squares regression, residuals, correlation

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: “Is the lunch line really slow?” Students guess, then do Part 1.
Day 2	Part 2: normal model and two-way table (or box plots for CP2).
Day 3	Part 3: regression and the data report; gallery walk of recommendations.

Materials

- Student handout and Cafeteria Data Cards
- Graphing calculators or a statistics app
- Graph paper; optional: real wait times collected by the class with stopwatches

How the Three Versions Differ

Level	What changes
CP2	Dot plots, box plots, mean/median, IQR and outliers, simple two-way table percents, and a given regression line.
CP1	Full comparison with SD, outlier removal, normal model and z-scores, conditional distributions, and LSRL with residual and r^2.

Level	What changes
Honors	Transformations of data, normal percentiles, expected counts, LSRL by formula, and leverage and influence.

Watch For (Common Misconceptions)

- Finding the median without ordering the data.
- Including the median itself when finding Q1 and Q3 of an odd-sized data set.
- Reading P(senior | salad) when the question asks P(salad | senior).
- Interpreting r^2 as “the percent of points on the line.”

Extension Ideas

- Collect real wait times for a week and repeat the analysis.
- Compare the LSRL with and without the influential point using technology.
- Design a survey to find out why students choose each line.

Answer Key – CP2

Part 1: Two Lines, Two Plots

#	Task	Answer
1	Make a dot plot of each line's wait times on the same scale (0 to 26 minutes).	Grill: clustered 4–14 with one dot at 25; Salad: clustered 3–10.
2	Find the median and the mean wait for each line.	Grill: median 8, mean 140 ÷ 15 ≈ 9.33 min. Salad: median 6, mean 95 ÷ 15 ≈ 6.33 min.
3	Why is the Grill mean larger than its median?	The 25-minute wait pulls the mean up; the median is not affected by one extreme value.
4	Which line usually has the shorter wait? Use a number to support your answer.	Salad Bar: its median (6 min) is 2 minutes less than the Grill's (8 min).

Part 2: Box Plots and Outliers

#	Task	Answer
5	Find the five-number summary for each line.	Grill: 4, 6, 8, 11, 25. Salad: 3, 5, 6, 8, 10.
6	Find the IQR of each line.	Grill: 11 − 6 = 5 min. Salad: 8 − 5 = 3 min.
7	Use the 1.5 × IQR rule to check the Grill data for outliers.	Upper fence 11 + 1.5(5) = 18.5; 25 > 18.5, so 25 minutes is an outlier. Lower fence −1.5, so no low outliers.
8	Draw both box plots on one scale and compare them in one sentence.	The Salad Bar has a lower median and much less spread (IQR 3 vs 5, range 7 vs 21).

Part 3: Choices and Predictions

#	Task	Answer
9	What percent of ninth graders choose salad? What percent of seniors?	10 ÷ 80 = 12.5%; 26 ÷ 80 = 32.5%
10	Is lunch choice associated with grade? Explain.	Yes; seniors choose salad at a much higher rate (32.5% vs 12.5%).
11	A computer gives the line $\hat{y} = 1.5 + 0.8x$ for wait time y and line length x. Predict the wait when 9 people are in line.	$\hat{y} = 1.5 + 0.8(9) = 8.7$ minutes
12	What does the slope 0.8 mean here?	Each additional person in line adds about 0.8 minute to the predicted wait.

Reflection

#	Prompt	Look for
R1	Should the principal use the mean or the median to describe the Grill wait? Why?	The median, because the data are skewed right by the 25-minute outlier.

Answer Key – CP1

Part 1: Compare the Lines

#	Task	Answer
1	Make parallel box plots and give the five-number summary of each line.	Grill: 4, 6, 8, 11, 25. Salad: 3, 5, 6, 8, 10.
2	Compute the mean and standard deviation of each line (use technology).	Grill: $\bar{x} \approx 9.33$, $s \approx 5.09$ min. Salad: $\bar{x} \approx 6.33$, $s \approx 1.95$ min.
3	Identify any outliers with the 1.5 × IQR rule, then find the Grill's mean and SD without it.	25 is an outlier (fence 18.5). Without it: $\bar{x} \approx 8.21$, $s \approx 2.78$ min.
4	Write a comparison of the two distributions that addresses shape, center, spread, and outliers in context.	Grill is skewed right with an outlier at 25 min, median 8, IQR 5; Salad is roughly symmetric, median 6, IQR 3, no outliers. Salad waits are shorter and more consistent.

Part 2: Normal Model and Choices

#	Task	Answer
5	Find the z-score of a day with an average wait of 12 minutes and interpret it.	$z = (12 - 9) \div 2 = 1.5$; that day was 1.5 SD above the mean.
6	About what percent of days have an average wait over 12 minutes?	$P(z > 1.5) \approx 0.0668$, about 6.7% of days
7	Find the percent of seniors who choose salad and the percent of salad buyers who are seniors.	$P(\text{salad} \mid \text{senior}) = 26 \div 80 = 32.5\%$; $P(\text{senior} \mid \text{salad}) = 26 \div 36 \approx 72.2\%$
8	Use conditional relative frequencies to decide whether lunch choice and grade are associated.	Salad: 12.5% vs 32.5%; Grill: 37.5% vs 30%; Pizza: 50% vs 37.5%. The distributions differ, so there is an association.

Part 3: Model the Wait

#	Task	Answer
9	Make a scatterplot and describe the direction, form, and strength.	Positive, linear, very strong.
10	Find the least-squares regression line and the correlation r.	$\hat{y} \approx 1.46 + 0.779x$; $r \approx 0.994$
11	Find the residual for the student who waited 9 minutes with 10 people in line. Interpret it.	$\hat{y} \approx 9.25$; residual $\approx 9 - 9.25 = -0.25$; that student waited about 0.25 min less than predicted.
12	Interpret r^2 and explain why predicting the wait for 40 people would be risky.	$r^2 \approx 0.988$: about 98.8% of the variation in wait time is explained by the linear model. $x = 40$ is far outside 2–15 (extrapolation).

Reflection

#	Prompt	Look for
R1	Why does removing the outlier change the standard deviation so much more than the median?	SD uses every distance from the mean (squared), so one extreme value inflates it; the median depends only on the middle position.
R2	Does the strong correlation prove that more people cause longer waits? Explain.	Correlation alone does not prove causation, though here the mechanism is plausible; other factors (number of servers, menu) could matter.

Answer Key – Honors

Part 1: Distributions

#	Task	Answer
1	Give the five-number summaries, means, and SDs of both lines, and identify outliers.	Grill: 4, 6, 8, 11, 25; $\bar{x} \approx 9.33$, $s \approx 5.09$; 25 is an outlier (fence 18.5). Salad: 3, 5, 6, 8, 10; $\bar{x} \approx 6.33$, $s \approx 1.95$; no outliers.
2	The cafeteria adds 2 minutes to every Grill wait to account for paying. What happens to the median, IQR, mean, and SD?	Median 10, mean ≈ 11.33 (both +2); IQR 5 and SD ≈ 5.09 unchanged.
3	The times are converted to seconds. Find the new median and SD of the Salad Bar.	Median 360 s; SD ≈ 1.95 × 60 ≈ 117 s
4	Write a full comparison and argue which measure of center and spread the principal should use for each line.	Grill is skewed right with an outlier → median and IQR; Salad is roughly symmetric → mean and SD are fine. Salad waits are shorter and less variable.

Part 2: Normal Model and Tables

#	Task	Answer
5	Find the proportion of days with averages between 8 and 11 minutes.	$P(-0.5 < z < 1) \approx 0.533$
6	The principal wants to flag the slowest 10% of days. Find the cutoff.	$z \approx 1.2816 \rightarrow 9 + 1.2816(2) \approx 11.56$ minutes
7	Make segmented bar graphs of lunch choice by grade and find P(senior | salad).	Gr 9: 37.5% Grill, 12.5% Salad, 50% Pizza; Gr 12: 30%, 32.5%, 37.5%. P(senior | salad) = 26 ÷ 36 ≈ 72.2%.
8	If grade and choice were not associated, how many seniors would be expected to choose salad? Compare with the data.	80 × 36 ÷ 160 = 18 expected; 26 observed, so seniors choose salad more than expected.

Part 3: Regression and Influence

#	Task	Answer
9	Use $\bar{x} = 7.75$, $\bar{y} = 7.5$, $s_x \approx 4.367$, $s_y \approx 3.423$, and $r \approx 0.994$ to find the LSRL by formula.	$b = r(s_y \div s_x) \approx 0.779$; $a = \bar{y} - b\,\bar{x} \approx 7.5 - 0.779(7.75) \approx 1.46$; $\hat{y} \approx 1.46 + 0.779x$
10	Interpret the slope, the y-intercept, r^2, and the residual at x = 10 (y = 9).	Slope: +0.78 min per person; intercept: about 1.5 min with nobody in line (service time); $r^2 \approx 0.988$; residual ≈ −0.25 min.
11	A new point (30 people, 8 min) is added. Is it an outlier, a high-leverage point, or influential? Predict how it changes the slope and r.	High leverage (x far from the others) and influential: it lies far below the line ($\hat{y} \approx 24.8$), so the slope and r would both decrease.
12	Describe what a residual plot would need to look like for the linear model to be appropriate.	Random scatter around 0 with no curve or fan shape.

Reflection

#	Prompt	Look for
R1	Why does adding a constant change the center but not the spread?	Every value moves by the same amount, so the distances between values do not change.
R2	How could one unusual point mislead the principal about the line-length model?	A high-leverage point can pull the line toward itself, changing the slope and r and making predictions unreliable.